160V NPN HIGH VOLTAGE TRANSISTOR IN SOT223

Features

- BV_{CEO} > 160V
- BV_{EBO} > 6V
- I_C = 600mA Continuous Collector Current
- Low Saturation Voltage (150mV max @10mA)
- h_{FE} specified up to 50mA for a high gain hold up
- Complementary PNP Type: DZT5401
- **Totally Lead-Free & Fully RoHS Compliant (Notes 1 & 2)**
- **Halogen and Antimony Free. "Green" Device (Note 3)**
- **For automotive applications requiring specific change control (i.e. parts qualified to AEC-Q100/101/104/200, PPAP capable, and manufactured in IATF 16949 certified facilities), please contact us or your local Diodes representative. https://www.diodes.com/quality/product-definitions/**

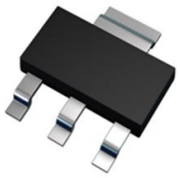
Mechanical Data

- Package: SOT223
- Package material: molded plastic. "Green" molding compound.
- UL Flammability Rating 94V-0
- Moisture Sensitivity: Level 1 per J-STD-020
- Terminals: Finish - Matte Tin Plated Leads, Solderable per MIL-STD-202, Method 208 e3
- Weight: 0.112 grams (approximate)

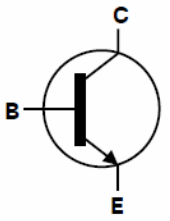
Applications

- High-voltage amplification applications
- High-voltage switching applications

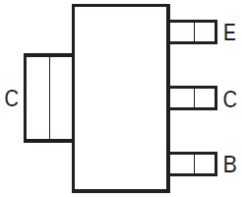
SOT223



Top View



Device Schematic



Pin-Out Top View

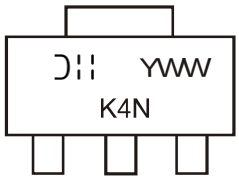
Ordering Information (Note 4)

Orderable Part Number	Package	Marking	Reel Size (Inches)	Tape Width (mm)	Packing	
					Quantity	Carrier
DZT5551-13	SOT223	K4N	13	12	2,500	Reel

Notes:
1. No purposely added lead. Fully EU Directive 2002/95/EC (RoHS), 2011/65/EU (RoHS 2) & 2015/863/EU (RoHS 3) compliant.
2. See https://www.diodes.com/quality/lead-free/ for more information about Diodes Incorporated's definitions of Halogen- and Antimony-free, "Green" and Lead-free.
3. Halogen- and Antimony-free "Green" products are defined as those which contain <900ppm bromine, <900ppm chlorine (<1500ppm total Br + Cl) and <1000ppm antimony compounds.
4. For packaging details, go to our website at https://www.diodes.com/design/support/packaging/diodes-packaging/.

Marking Information

(Top View)



K4N = Product type marking code
= Manufacturer's code marking
YWW = Date code marking
Y = Last digit of year ex: 7 = 2007
WW = Week code 01 - 52

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Maximum Ratings (@T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Value	Unit
Collector-Base Voltage	V_{CBO}	180	V
Collector-Emitter Voltage	V_{CEO}	160	V
Emitter-Base Voltage	V_{EBO}	6	V
Continuous Collector Current	I_C	600	mA
Peak Collector Current	I_{CM}	1	A

Thermal Characteristics

Characteristic	Symbol	Value	Unit
Power Dissipation (Note 5)	P_D	2	W
Thermal Resistance, Junction to Ambient (Note 5)	$R_{\theta JA}$	62.5	°C/W
Thermal Resistance, Junction to Leads (Note 6)	$R_{\theta JL}$	45	°C/W
Thermal Resistance, Junction to Case (Note 7)	$R_{\theta JC}$	27	°C/W
Operating and Storage Temperature Range	T_J, T_{STG}	-55 to +150	°C

Notes: 5. Device mounted on 50mm X 50mm X 1.6mm FR-4 PCB with high coverage of single sided 1 oz. copper, in still air condition
6. Thermal resistance from junction to solder-point (at the end of the collector lead).
7. Thermal resistance from junction to the top of the case.

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Thermal Characteristics and Derating Information

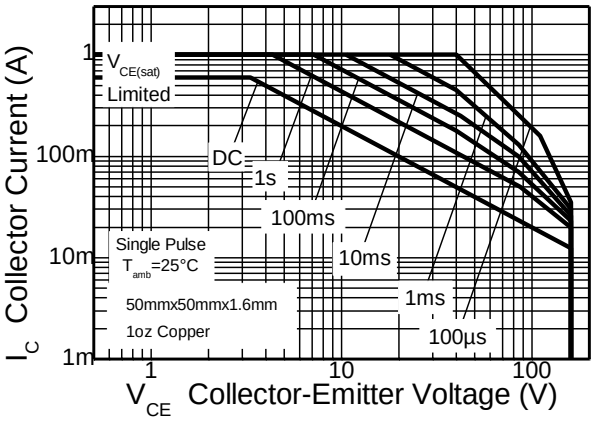

Figure 1. Safe Operating Area

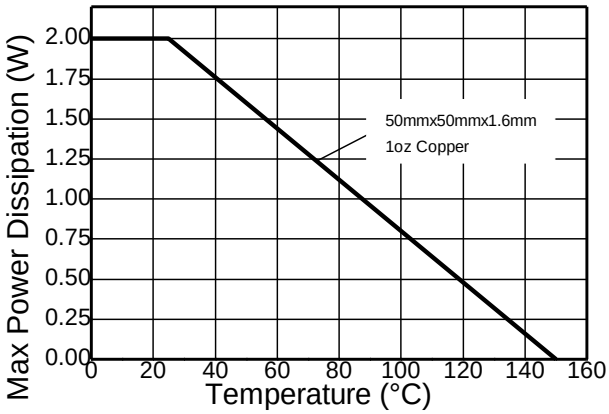

Figure 2. Derating Curve

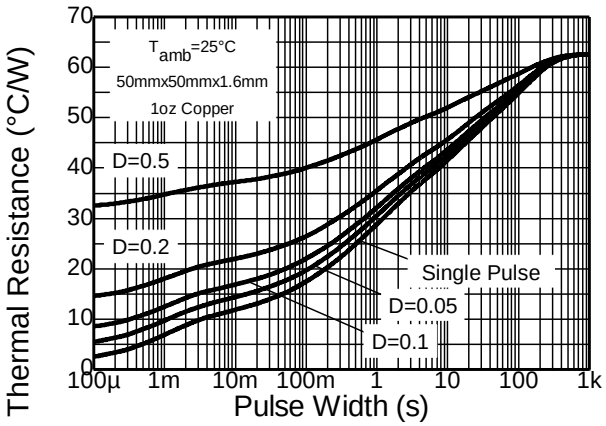

Figure 3. Transient Thermal Impedance

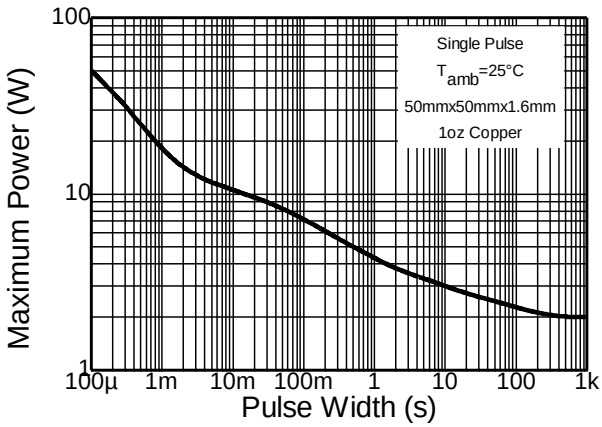

Figure 4. Pulse Power Dissipation

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Electrical Characteristics (@T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Min	Typ	Max	Unit	Test Condition
OFF CHARACTERISTICS						
Collector-Base Breakdown Voltage	BV_{CBO}	180	270	—	V	I_C = 100µA
Collector-Emitter Breakdown Voltage (Note 8)	BV_{CEO}	160	200	—	V	I_C = 1mA
Emitter-Base Breakdown Voltage	BV_{EBO}	6.0	7.85	—	V	I_E = 100µA
Collector Cutoff Current	I_{CBO}	— —	1 —	50 50	nA µA	V_{CB} = 120V V_{CB} = 120V, T_A = +100°C
Emitter Cutoff Current	I_{EBO}	—	1	50	nA	V_{EB} = 4V
ON CHARACTERISTICS (Note 8)						
Collector-Emitter Saturation Voltage	$V_{CE(sat)}$	— —	65 115	150 200	mV mV	I_C = 10mA, I_B = 1mA I_C = 50mA, I_B = 5mA
Base-Emitter Saturation Voltage	$V_{BE(sat)}$	— —	760 840	1000 1200	mV mV	I_C = 10mA, I_B = 1mA I_C = 50mA, I_B = 5mA
DC Current Gain	h_{FE}	80 80 30	130 145 65	— 250 —	—	I_C = 1mA, V_{CE} = 5V I_C = 10mA, V_{CE} = 5V I_C = 50mA, V_{CE} = 5V
SMALL SIGNAL CHARACTERISTICS						
Transition Frequency	f_T	100	130	300	MHz	V_{CE} = 10V, I_C = 10mA, f = 100MHz
Small Signal Current Gain	h_{fe}	50	—	260	—	V_{CE} = 10V, I_C = 10mA, f = 1kHz
Output Capacitance	C_{obo}	—	—	6	pF	V_{CB} = 10V, f = 1MHz
Noise Figure	NF	—	—	8	dB	V_{CE} = 5.0V, I_C = 200µA, R_S = 1.0kΩ, f = 1.0kHz
Delay Time	$t_{(d)}$	—	95	—	ns	V_{CC} = 10V, I_C = 10mA, I_{B1} = -I_{B2} = 1mA
Rise Time	$t_{(r)}$	—	64	—	ns	
Storage Time	$t_{(s)}$	—	1256	—	ns	
Delay Time	$t_{(f)}$	—	140	—	ns	

Notes: 8. Pulse Test: Pulse width ≤ 300µs. Duty cycle ≤ 2.0%.

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Typical Electrical Characteristics (@T_A = +25°C, unless otherwise specified.)

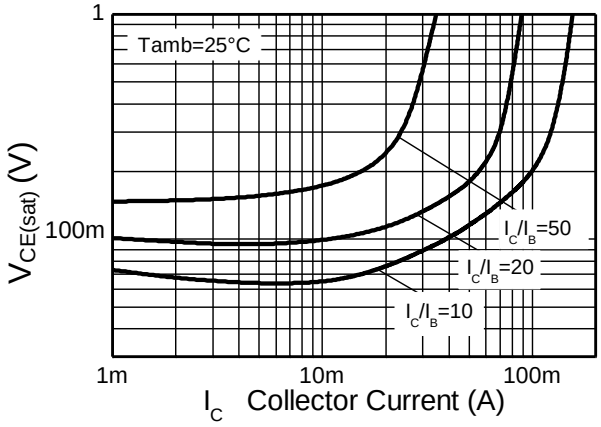

Figure 5. $V_{CE(sat)}$ v I_C

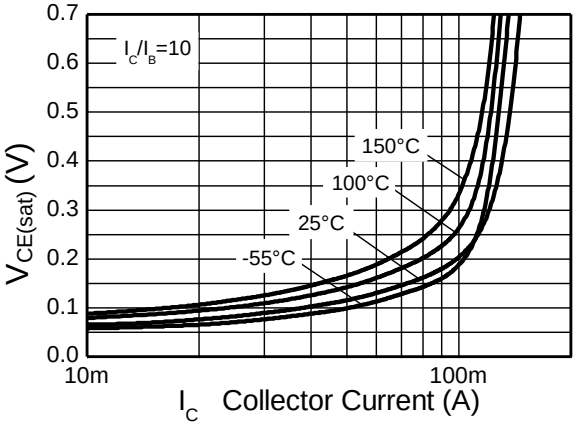

Figure 6. $V_{CE(sat)}$ v I_C

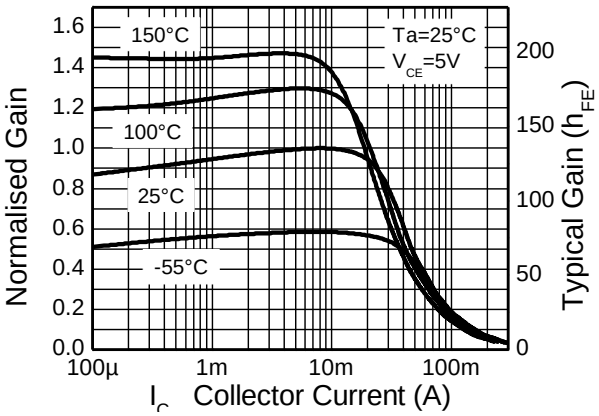

Figure 7. h_{FE} v I_C

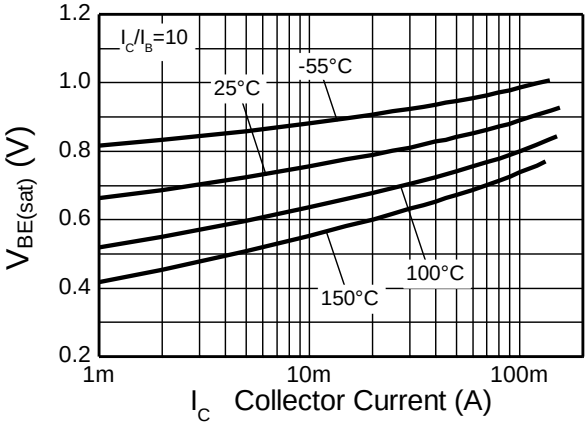

Figure 8. $V_{BE(sat)}$ v I_C

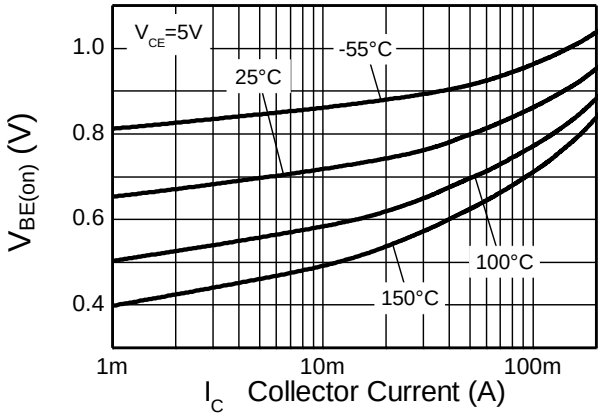

Figure 9. $V_{BE(on)}$ v I_C

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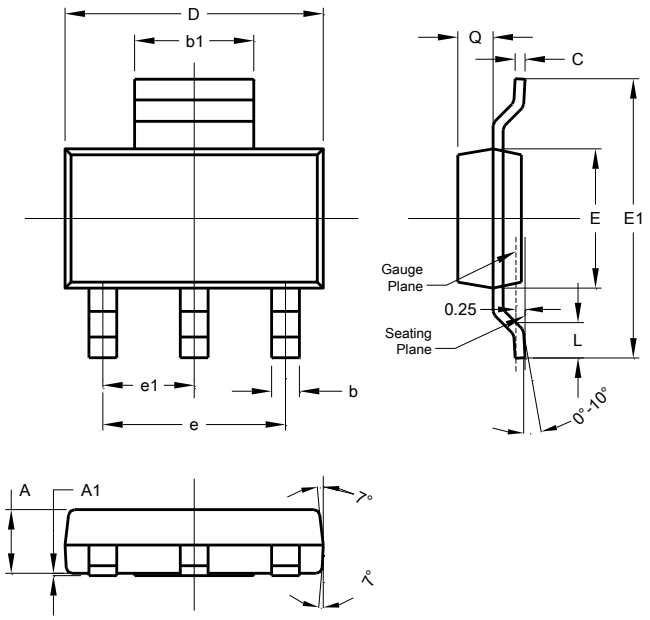
Package Outline Dimensions

Please see http://www.diodes.com/package-outlines.html for the latest version.

SOT223

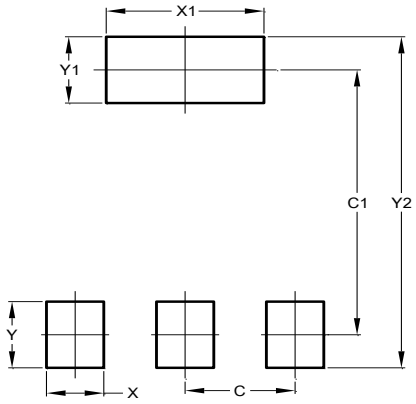


SOT223			
Dim	**Min**	**Max**	**Typ**
A	1.55	1.65	1.60
A1	0.010	0.15	0.05
b	0.60	0.80	0.70
b1	2.90	3.10	3.00
C	0.20	0.30	0.25
D	6.45	6.55	6.50
E	3.45	3.55	3.50
E1	6.90	7.10	7.00
e	-	-	4.60
e1	-	-	2.30
L	0.85	1.05	0.95
Q	0.84	0.94	0.89
All Dimensions in mm			

Suggested Pad Layout

Please see http://www.diodes.com/package-outlines.html for the latest version.

SOT223



Dimensions	Value (in mm)
C	2.30
C1	6.40
X	1.20
X1	3.30
Y	1.60
Y1	1.60
Y2	8.00

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